

# **The Core Ontology of Reality: Evolutionary Access, Planck-Scale Discreteness, and the Quest for Unification**

## **Introduction**

The pursuit of a unified theory of physical reality has historically relied upon the continuous expansion of mathematical formalisms, generating increasingly intricate structures to encapsulate the behavior of phenomena at extreme scales. From the tensor calculus mapping the manifold of General Relativity to the infinite-dimensional Hilbert spaces foundational to Quantum Field Theory, the trajectory of modern physics has been characterized by structural accretion. However, an alternative epistemological and ontological paradigm suggests that the bedrock of physical reality is not characterized by labyrinthine continuous mathematics, but by a "Core Ontology"—an irreducible, minimal set of logical and structural primitives from which all observable physical laws deterministically or probabilistically emerge.<sup>1</sup>

This report explores the hypothesis that a unified framework of physics is not merely a distant theoretical ambition but an already instantiated reality, physically realized within the technological artifacts of human civilization and cognitively embedded in the evolutionary architecture of the human mind.<sup>4</sup> By integrating evolutionary epistemology, digital physics, and the rigorous ontological deconstruction of major physical theories, the ensuing analysis reveals that continuous spacetime and fundamental integers are emergent, macroscopic properties rather than foundational truths.<sup>6</sup> At the Planck scale, the universe operates as a profoundly dense, discrete informational substrate.<sup>8</sup>

From this perspective, the theoretical convergence of quantum mechanics, relativity, thermodynamics, and information theory strongly implies the existence of an ultra-compact, self-referential "reality generator".<sup>10</sup> The analysis presented herein systematically unpacks the evolutionary, physical, and computational evidence for this generative core, culminating in a concrete research directive to map the deepest isomorphisms of natural law and deconstruct contemporary physics to its axiomatic skeleton.

## **1. Grounding the Inquiry: A Mind Experiment in Evolutionary Epistemology**

### **The Phylogenesis of Cognition and Ontological Access**

To comprehend the architecture of physical reality, it is first necessary to examine the cognitive

apparatus through which that reality is apprehended and modeled. Evolutionary epistemology posits that the cognitive capacities of *Homo sapiens* are not arbitrary historical accidents but the highly optimized products of natural selection interacting with the fundamental structure of the environment.<sup>1</sup> The phylogenesis of cognition is an extended process wherein experience, acting against the background of genetic variation, stochastically shapes the development of a species by selecting for an individual's ability to accurately map and predict the environment.<sup>4</sup> If the universe were fundamentally chaotic, or if it lacked a cohesive, computationally reducible underlying logic, the emergence of a biological cognitive system capable of mathematical abstraction would be probabilistically impossible. Instead, the human mind has been sculpted by evolutionary pressures to intuitively apprehend the Core Ontology of the universe.

The discrepancy between formal axiomatic logic and human intuitive mathematics serves as a primary indicator of this evolutionary optimization. In the foundational mathematical text *Principia Mathematica*, Alfred North Whitehead and Bertrand Russell required hundreds of pages of rigorous logical definitions, axioms, and lemmas to formally establish the simple arithmetic proposition  $1 + 1 = 2$ .<sup>13</sup> Furthermore, establishing this identity required complex set-theoretic scaffolding intended to reduce all classical mathematics to pure logic.<sup>14</sup> Yet, the human mind grasps this relation instantaneously, often before formal schooling begins. This is not a trivial cognitive shortcut; it demonstrates that evolutionary processes have already performed the ontological heavy lifting.<sup>4</sup> Natural selection has acted as a multi-millennial algorithmic filter, embedding a highly compressed, intuitive representation of physical laws and logical relations directly into human neural circuitry.

As organisms navigate their physical environment, their survival depends upon their ability to accurately predict outcomes based on physical interactions. Biological perception itself is the core ontology under examination when inquiring into learning and cognition; organisms discover perceptual structures that enable them to coordinate actions within a socio-material context.<sup>15</sup> Consequently, the human capacity for mathematics is an emergent readout of the universe's own structural logic. The universe is computable, and the biological brain evolved as a simulator capable of internalizing that computation.<sup>10</sup> This alignment is further explored in cybernetics, where self-reference and information theory are seen as deeply tied to evolutionary epistemology, demonstrating that cognitive systems mirror the recursive feedback loops present in physical laws.<sup>16</sup>

## **The Illusion of the Fundamental Integer and the Planck-Scale**

Despite this intuitive cognitive access, the progressive refinement of human empirical inquiry over five millennia—from Babylonian arithmetic and Euclidean geometry to Newtonian mechanics and the quantum revolution—has exposed the strict limits of macroscopic biological intuition. The most profound realization of this cumulative scientific endeavor is that the intuitive integer "1," while cognitively necessary for macroscopic survival and object permanence, completely dissolves at the foundational limits of physical reality.<sup>6</sup>

When examining reality at the absolute minimal resolvable scales, the smooth continuous

mathematics of classical physics gives way to severe discrete limits. The Planck length:

$$l_p = \sqrt{\frac{\hbar G}{c^3}} \approx 1.616 \times 10^{-35} \text{ m}$$

and the Planck time:

$$t_p = \sqrt{\frac{\hbar G}{c^5}} \approx 5.391 \times 10^{-44} \text{ s}$$

represent the fundamental bounds of observable space and chronometric duration. At this scale, the concept of a context-independent, isolated unity breaks down.<sup>6</sup> What human cognition registers as discrete macroscopic particles, continuous fields, or even exact integers are emergent patterns arising from an unimaginably dense web of discrete, sub-Planckian informational interactions.<sup>6</sup>

The physicist John Archibald Wheeler captured this paradigm shift with his "It from Bit" proposition, arguing that the universe is fundamentally informational, built from discrete quantum events or binary choices.<sup>6</sup> In this view, the universe at its core is a vast quantum computer evaluating Boolean functions that represent the laws of physics.<sup>7</sup> Stabilizing the macroscopic concept of unity—such as identifying a single, persistent electron or a localized atom—requires an astronomically large sequence of Planck-scale informational exchanges. The integer "1" is therefore not a physical primitive; it is a statistical summary of a profoundly complex computational process. The fact that the human brain effortlessly utilizes the concept of unity without consciously processing the underlying Planck-scale events highlights the brain's role as an ultimate data-compression engine, honed by evolution to navigate the Core Ontology without being computationally overwhelmed.<sup>4</sup>

## 2. The Technological Instantiation of Unification

A persistent narrative within theoretical physics is that the ultimate unification of General Relativity (describing macro-scale gravity and the cosmos) and Quantum Mechanics (describing micro-scale interactions) is a distant, unrealized academic grail.<sup>20</sup> However, an application of the Core Ontology framework yields a highly provocative counter-claim: a unified theory of physics has already been achieved in practice, embodied directly within the technological artifacts of human civilization.<sup>22</sup>

### The Microprocessor as a Unified Physical System

The modern silicon microprocessor represents the physical instantiation of a practically unified theory. Contemporary microprocessors contain billions of transistors switching at extreme frequencies. The operational stability and flawless execution of these devices cannot be explained by any single, isolated physical framework; their function requires the simultaneous, harmonious integration of multiple fundamental physical domains.<sup>5</sup>

At the nanometer scale of modern transistor gates, the wave-like nature of electrons

necessitates a strict accounting for quantum tunneling. The probabilities of electron transfer across potential barriers govern leakage currents and discrete switching states, meaning quantum mechanics is not merely a theoretical abstraction but a daily engineering constraint.<sup>5</sup> Simultaneously, the synchronization of global networks and satellite constellations (such as GPS) that feed data to these terrestrial microprocessors requires constant relativistic corrections.<sup>25</sup> Time dilation due to relative velocity (Special Relativity) and differences in gravitational potential (General Relativity) must be integrated seamlessly for exact chronological precision across the computational network.

Furthermore, the operation of these systems is bound by the laws of thermodynamics. The erasure of information within logic gates involves mandatory entropy increases, governed by Landauer's principle. The extraction of dissipated heat from tightly packed nanoscale architectures is a purely thermodynamic constraint that dictates the physical layout of the processing cores.<sup>5</sup> Overarching all of these physical realities is Information Theory, which dictates the strict limits of Shannon entropy, error-correction rates, and algorithmic complexity.<sup>10</sup>

## **The Generative Substrate Overcoming Theoretical Friction**

No single unified mathematical equation currently encompasses all these regimes in theoretical physics.<sup>20</sup> Yet, the physical silicon chip functions seamlessly. This operational reality suggests that nature does not experience a conflict between relativistic gravity, quantum probabilities, and thermodynamic entropy. The perceived incompatibility lies entirely within human mathematical formalisms—specifically, the theoretical friction between continuous differential equations (used in gravity) and discrete state vectors (used in quantum mechanics).<sup>8</sup>

The technological success of the microprocessor demonstrates that a shared generative substrate inherently unites these domains. The unification is occurring within the silicon itself, processed by the Core Ontology of the universe. This proves that nature's foundational ruleset is intrinsically cohesive, scale-invariant, and capable of simultaneously running quantum, relativistic, and thermodynamic "software" on the same discrete physical "hardware." Thus, the unification of physics resides not in a yet-to-be-discovered tensor equation, but in the underlying generative algorithm that all successful theories have been unconsciously mapping.<sup>21</sup>

## **3. Ontological Deconstruction of Major Physical Frameworks**

To explicitly identify the generative substrate that microprocessors unconsciously map, contemporary physics must undergo a rigorous ontological archaeology. This requires systematically stripping the major theoretical frameworks of their historical, continuous, and phenomenological baggage to reveal their minimal axiomatic skeletons.<sup>28</sup>

### **3.1 General Relativity: From Manifolds to Causal Information**

## Gradients

In its standard geometric formulation, General Relativity models gravity as the continuous curvature of a four-dimensional spacetime manifold, quantified by the Einstein field equations.<sup>20</sup> However, an ontological reduction demands moving beyond continuous geometry. Theories such as Causal Set Theory and recent formulations of digital physics suggest that spacetime is fundamentally discrete, quantized in Planck units, and "qubitised".<sup>7</sup> In this reduced ontology, spacetime is not a smooth mathematical manifold but a causal partial order defined exclusively on discrete computational events.<sup>30</sup>

The continuous "curvature" of spacetime is therefore reinterpreted as a coarse-grained, macroscopic description of underlying informational gradients.<sup>30</sup> The geometry of the universe emerges from the density of these discrete causal interactions. When massive objects are present, the density of causal links increases, modulating the flow of proper time and resulting in the macroscopic phenomenon identified as gravity.<sup>31</sup> Thus, the Einstein field equations are not the ultimate base reality; they are a fluid-dynamic, statistical approximation of an underlying discrete causal network.

## 3.2 Quantum Mechanics: Informational Axioms and Bundles of Properties

The traditional formulation of Quantum Mechanics relies on complex Hilbert spaces, wave functions, and measurement collapse postulates, which introduce significant interpretive and ontological difficulties.<sup>8</sup> Ontological deconstruction shifts the focus entirely from mechanical states to informational primitives.

Researchers in quantum foundations, such as Giacomo Mauro D'Ariano, Giulio Chiribella, and Paolo Perinotti, have demonstrated that the entirety of quantum theory can be derived from pure informational principles without any reliance on classical mechanical primitives.<sup>3</sup> Their framework identifies minimal axioms—such as causality, local distinguishability, and pure steering—that treat reality fundamentally as an information processing system. By employing quantum cellular automata and imposing strict, simple topological constraints (locality, unitarity, homogeneity, and isotropy), complex relativistic equations such as the Dirac and Weyl equations emerge naturally without pre-supposing a continuous spacetime background.<sup>29</sup> This effectively provides a solution to Hilbert's sixth problem by mathematizing physics without physical primitives, relying solely on mathematical algorithms.<sup>29</sup>

Simultaneously, philosophical investigations into quantum ontology, such as those conducted by Olimpia Lombardi and Sebastian Fortin, highlight the necessity of abandoning standard "individual-ontology" in physics.<sup>36</sup> By contrasting the "stuff-ontology" of chemistry with the "individual-ontology" of physics, they propose a modal-Hamiltonian interpretation where quantum systems are not localized particles but rather "non-individual bundles of possible properties".<sup>37</sup> These bundles are defined strictly by symmetry groups (such as the Galilei group for non-relativistic mechanics and the Poincaré group for relativistic mechanics).<sup>37</sup> In this view,

the classical "particle" is an illusion generated by the intersection of these properties under specific measurement conditions. The wave function is not a fundamental physical object but a statistical summary of an underlying generative process characterized by informational conservation.<sup>37</sup>

### 3.3 String Theory and Duality: The Common Core Theory

String Theory operates in 10 or 26 dimensions, substituting point particles with extended vibrating strings to reconcile gravity and quantum mechanics.<sup>20</sup> While mathematically profound, its high dimensionality has frustrated empirical validation. Ontologically, these extra dimensions can be understood not as literal physical spaces, but as higher-order mathematical approximations of the internal state space of the Core Ontology's discrete generator.

The concept of physical duality (such as T-duality or the AdS/CFT correspondence) is critical to deconstructing String Theory. When two distinct physical models describe the identical physical reality, they point to the existence of a "common core theory".<sup>39</sup> Through abstraction, the "additional structure" unique to each specific model is removed, leaving only the structure that is isomorphic between them.<sup>39</sup> This common core theory captures the formal essence of the physics and represents the "safe choice" for scientific realism.<sup>39</sup> String theory's critical dimensions are therefore perturbative expansions around this true minimal description, representing the mathematical degrees of freedom necessary to resolve the discrete generator's operations using human continuous mathematics.

### 3.4 Thermodynamics and Information Theory: Erasure and Compression

Thermodynamics is universally characterized by the Second Law—the inexorable increase in macroscopic entropy over time. In the Core Ontology framework, Thermodynamics and Information Theory are not distinct disciplines; they are synonymous macroscopic read-outs of the universe's computational efficiency.<sup>26</sup>

Information Theory measures algorithmic complexity and Shannon entropy. When a physical system evolves, it processes information.<sup>10</sup> The Second Law of Thermodynamics acts as the statistical shadow of irreversible information erasure within a finite-memory universe.<sup>26</sup> Just as a digital computer generates measurable heat when clearing bits due to Landauer's principle, the universe generates macroscopic entropy as its discrete generator overwrites local state histories to compute the subsequent global state. Consequently, thermodynamic laws are not independent driving forces; they are the unavoidable constraints of algorithmic computation applied to a finite physical substrate.

### Summary of Ontological Deconstruction

To synthesize the preceding analysis, the following table maps the traditional macroscopic phenomenological frameworks to their deconstructed, generative primitives within the Core Ontology:

| Traditional Theoretical Framework | Macroscopic Phenomenological View            | Deconstructed Generative Primitive (Core Ontology)                                                          |
|-----------------------------------|----------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| <b>General Relativity</b>         | Continuous 4D spacetime curvature.           | Causal partial order on discrete events; informational density gradients. <sup>28</sup>                     |
| <b>Quantum Mechanics</b>          | Wave functions; distinct point particles.    | Isotropic informational updating; non-individual bundles of possible properties. <sup>29</sup>              |
| <b>Classical Mechanics</b>        | Deterministic trajectories and solid bodies. | Thermodynamic limit of coarse-grained state transitions; minimum change optimization. <sup>41</sup>         |
| <b>Information Theory</b>         | Shannon entropy; channel capacity limits.    | Direct read-outs of the reality generator's data compression efficiency. <sup>10</sup>                      |
| <b>String Theory</b>              | Multi-dimensional vibrating strings/branes.  | Mathematical approximations of the internal state space; isomorphic "common core" structures. <sup>39</sup> |
| <b>Thermodynamics</b>             | Inexorable entropy increase (Arrow of Time). | Statistical shadow of irreversible information erasure within a finite memory substrate. <sup>26</sup>      |

## 4. The Reality Generator Hypothesis

The theoretical convergence of these deconstructed ontologies strongly supports the existence of an underlying, unified algorithmic substrate. This is formally termed the *Reality Generator* hypothesis. Inspired by advances in digital physics, quantum cellular automata, and hypergraph rewriting systems (such as Stephen Wolfram's computational universe and the "Ruliad" concept, or Richard Lipton's unique binary system), this hypothesis posits that the entirety of physical law is the direct output of an ultra-minimal, self-referential logical



structure.<sup>2</sup>

## **Definitional Properties of the Generative Core**

To successfully underwrite all physical phenomena without falling into the philosophical trap of infinite regress, the reality generator must exhibit four essential properties:

### **1. Extreme Compactness and Algorithmic Parsimony**

The base code of the universe cannot be infinite or arbitrarily complex. It must be representable as an ultra-compact set of generative rules that conceptually fit within a volume far smaller than the Planck length.<sup>11</sup> Much like a minimal fractal equation generates infinite, non-repeating complexity, the core algorithm of the universe requires only foundational informational instructions.<sup>2</sup> The vast complexity of the observable cosmos is not the result of complex rules, but the consequence of simple rules iterated an astronomically large number of times over deep time.<sup>2</sup> In this context, fundamental physical constants (such as the fine-structure constant or the gravitational constant) are not arbitrary inputs requiring a fine-tuning mechanism; they are emergent, stable fixed points of the generator's internal attractors.<sup>46</sup>

### **2. Self-Reference and Recursive Feedback**

For a system to generate stable, complex structures like matter and energy, its rules must operate continuously upon their own outputs. This self-reference naturally yields the fractal topologies, recursive hierarchies, and power-law statistics observed across all scientific disciplines—from the spatial distribution of galactic clusters to Zipf's law in linguistics and the Mandelbrot sets seen in biological and geological formations. In Quantum Mechanics, phenomena such as superposition and entanglement can be understood as the self-referential reuse of generative rules across differing topological scales, where distinct physical locations remain logically linked within the generator's state matrix because they share the same computational lineage.<sup>36</sup>

### **3. Scale Invariance with Graceful Degradation**

A true unified generator must operate seamlessly from the sub-Planckian quantum foam to the cosmological horizon. The perceived differences between quantum and classical mechanics are merely artifacts of the observer's coarse-graining.<sup>48</sup> At the Planck scale, the discrete jumps of the generator are visible, manifesting as quantum phenomena. As these events are aggregated and averaged out at human scales, the discrete steps blur into smooth probabilities and classical trajectories. This represents a "graceful degradation" into Classical Mechanics and General Relativity, where the principle of minimum change (or "laziness of nature") governs macroscopic motion.<sup>41</sup>

### **4. The Resolution of Gödelian Incomputability**

A persistent philosophical critique of digital physics is the argument derived from Gödel's Incompleteness Theorems and Turing uncomputability. Critics argue that classical continuous



physical laws—such as the uncountably infinite degrees of freedom in a quantum wavefunction or the nonlinear dynamics of partial differential equations like the Navier-Stokes equations—cannot be perfectly simulated by a finite Turing machine.<sup>49</sup>

The Reality Generator hypothesis elegantly circumvents this critique by positing that the universe itself *is not continuous*. The apparent incomputability lies entirely within human mathematical abstractions (the continuum, infinity, uncountability), not in the physical reality itself. If reality operates on a finite, discrete, quantum-computational substrate at the Planck scale, it avoids the mathematical infinities that plague continuous theoretical physics.<sup>6</sup> The universe does not "compute" real numbers with infinite decimal expansions; it executes finite, discrete state updates that our macroscopic mathematics can only approximate with continuous functions.<sup>6</sup> Thus, Gödelian limits apply to the formal axiomatic systems created by human minds, not to the discrete algorithmic execution of the universe.<sup>50</sup>

## 5. The Research Imperative: Ontological Archaeology

If the Core Ontology is already intuitively embedded in human cognition and operationally functional in human technology, the frontier of physics shifts fundamentally. The primary obstacle to theoretical unification is no longer experimental energy limits (which require prohibitively massive particle colliders), but epistemological and mathematical fragmentation. The proposition of higher-dimensional string vacuums will only yield more phenomenological mathematical data; it will not reveal the core computational architecture.

The path forward requires the establishment of a systematic discipline of *ontological archaeology*—a rigorous program designed to strip away macroscopic mathematical approximations and isolate the reality generator. This imperative involves a coordinated, interdisciplinary approach spanning theoretical physics, information theory, and computer science.<sup>22</sup>

### Objectives of the Ontological Archaeology Program

The systematic deconstruction of physics requires four sequential objectives to test for convergence on the generative core:

1. **Catalog the Minimal Axiom Sets:** Theoretical physicists and mathematicians must systematically isolate the absolute minimum axioms required to sustain every empirically successful theory. As demonstrated by D'Ariano, Chiribella, and Perinotti for quantum theory, the goal is to define frameworks not by their moving parts (e.g., continuous fields, point particles), but by their strict informational and logical constraints (e.g., causality, reversibility, localized distinguishability, homogeneity).<sup>3</sup>
2. **Reduce Frameworks to Generative Rules:** Once the minimal axioms are isolated, they must be translated into discrete generative rules. For instance, the continuous differential equations of fluid dynamics and spacetime metrics must be rewritten in the language of quantum cellular automata, causal sets, or hypergraph rewrites.<sup>28</sup> The objective is to find the minimum algorithmic instruction capable of outputting the specific physical axiom.

3. **Search for Isomorphisms and Fixed Points:** With physical theories reduced to generative rules, the next step is rigorous cross-disciplinary mapping. Utilizing advanced, AI-assisted automated theorem provers and machine learning algorithms<sup>22</sup>, researchers must scan these massive rule sets for isomorphisms—structural similarities where two seemingly different physical laws share the identical algorithmic root. Just as the "common core theory" links dual string models by extracting shared logical syntax<sup>39</sup>, this search will computationally identify the mathematical intersections between quantum unitarity and thermodynamic entropy.
4. **Synthesize and Test the Generator:** The final objective is to synthesize these isolated isomorphisms into a single, cohesive, self-referential generator. This entails proposing specific algorithmic codes or topological logical expressions. These candidate generators can then be executed via high-performance computing (HPC) or advanced quantum simulations.<sup>20</sup> The ultimate test is to determine if their low-order computational iterations successfully recover the Standard Model of particle physics, while their macroscopic emergent properties recover General Relativity and the Second Law of Thermodynamics simultaneously.

Success in this endeavor would mean that the ultimate "Theory of Everything" will not be a multi-volume textbook of dense Lagrangians and intractable continuous equations. Instead, it will be a profoundly elegant, compact few lines of code or logical syntax.

## Conclusion

The pursuit of unification in theoretical physics has reached a pivotal epistemological and ontological threshold. Through a rigorous cognitive archaeology of the human species, it is evident that evolutionary processes have bequeathed *Homo sapiens* a privileged, intuitive access to the logical foundations of nature. The instantaneous comprehension of abstract mathematics and systemic relations points directly toward a universe governed by computable, algorithmic rules rather than incalculable chaos.

However, the continued reliance on continuous mathematics and the concept of an indivisible integer "1" has obscured the discrete, sub-Planckian informational substrate that truly governs physical reality. By acknowledging that continuity is an emergent macroscopic illusion, the apparent theoretical incompatibilities between General Relativity, Quantum Mechanics, and Thermodynamics completely dissolve. This unification is not merely theoretical; it is practically demonstrated billions of times a second in the synchronized, multi-domain operations of modern silicon microprocessors.

By deconstructing established physical frameworks to their primitive ontological cores, a singular necessity emerges: an ultra-compact, self-referential, scale-invariant reality generator. The universe is not a collection of independent substances and forces floating in an empty continuum; it is a unified informational process executing an exquisitely simple generative rule across an unfathomable expanse of discrete computational events.

The mandate for the global scientific community is therefore clear. The era of adding ad hoc

dimensions and continuous parameters must transition into an era of rigorous ontological reduction and algorithmic archaeology. By systematically distilling contemporary physics into its minimal axiomatic components and leveraging advanced computational resources to map their intersections, humanity can isolate the generative core of reality. The key to the universe has been forged by evolution over deep time and turned by centuries of scientific inquiry; the final task is to map, reduce, and recognize the underlying code that has been silently generating both the cosmos and the human capacity to comprehend it.

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